This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Secure Event-Based Consensus Control for Multi-Agent Systems Under DoS Attacks and Input Saturation

Yifang Zhang, Zheng-Guang Wu, Ka-Wai Kwok, *Senior Member, IEEE*, Tingwen Huang, Prasun Chakrabarti, and Le Zhou, *Senior Member, IEEE*

***Abstract*—The secure consensus problem is addressed for multiagent systems (MASs) suffering from saturated control input and denial-of-service (DoS) attacks. The communication networks' open setting and sharing nature give rise to security issues and impact the performance of MASs. Malicious DoS attacks attempt to disrupt the information exchange and undermine consensus by compromising the availability of transmitted data. Moreover, the control input can be saturated as a result of physical device limitations or safety concerns. To tackle these challenges, a state-prediction-based dynamic event-triggered mechanism (DETM) control protocol is designed to guarantee the secure consensus of MASs while reducing redundant communication, avoiding continuous monitoring of adjacent states, and ensuring effective utilization of limited bandwidth resources. Zeno behavior is eliminated by confirming the existence of a positive lower bound on interevent intervals. Sufficient conditions are established for the co-design of the DETM and controller to accomplish the desired goal. Finally, a simulation is conducted to substantiate the effectiveness and validity of the proposed control protocol.**

***Index Terms*—Consensus, denial-of-service (DoS) attacks, dynamic event-triggered mechanism (DETM), multiagent system (MAS), saturated control input.**

Received 20 December 2024; accepted 1 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62233015 and Grant 62173306; in part by the Research Grants Council (RGC) of Hong Kong under Grant 17210023, Grant C4026-21G, and Grant STG1/E-401/23-N; and in part by the Centre for Garment Production Limited and Multi-scale Medical Robotics Center Limited, both funded by Innovation and Technology Commission. This article was recommended by Associate Editor H. Zhang. *(Corresponding author: Le Zhou.)*

Yifang Zhang and Zheng-Guang Wu are with the State Key Laboratory of Industrial Control Technology, Institute of Cyber-Systems and Control, Zhejiang University, Hangzhou 310027, China (e-mail: yifangzhang@zju.edu.cn; nashwzhg@zju.edu.cn).

Ka-Wai Kwok is with the Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Hong Kong (e-mail: kwokkw@mae.cuhk.edu.hk).

Tingwen Huang is with Science Program, Texas A& M University at Qatar, Doha, Qatar (e-mail: tingwen.huang@qatar.tamu.edu).

Prasun Chakrabarti is with the Techno India NJR Institute of Technology, Udaipur 313003, India, and also with ITM SLS Baroda University, Vadodara 391510, India (e-mail: drprasun.cse@gmail.com).

Le Zhou is with the School of Automation and Electrical Engineering, Zhejiang University of Science and Technology, Hangzhou 310023, Zhejiang, China (e-mail: zhoule@zust.edu.cn).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TSMC.2025.3558286.

Digital Object Identifier 10.1109/TSMC.2025.3558286

I. Introduction

MULTIAGENT systems (MASs), which are integrated with distributed control, network communication and sensing, are usually employed to deal with elaborate multitasks cooperatively. Consensus of MASs is a fundamental challenge, requiring multiple agents, such as autonomous robots or drones, to achieve a shared understanding or agreement on a specific state over time through local interactions. The consensus problem has garnered significant attention owing to its extensive applicability across various domains, including autonomous surface vehicles [1], wireless sensor networks [2], smart grid systems [3], and mobile train lifting jack systems [4]. A basic issue for consensus problem lies in identifying an appropriate control protocol that can determine the data transmission between agents via the open and sharing communication networks [5].

The security of MASs has become increasingly critical due to the vulnerabilities arising from complex interactions and inherent openness of communication networks. Given the distributed display of the agents, the quality of information transmission is essential in tackling consensus problems. Nevertheless, the reliability of communication networks is susceptible to various cyber attacks, such as denial-of-service (DoS) attacks, which can lead to information losses between agents and wreak havoc on overall system performance [6]. DoS attacks, which can be readily initiated within communication networks, attempt to block communication channels. Even though malicious adversaries have limited prior knowledge about the controlled systems, they can still lead to packet drop and even consensus failure, resulting in significant damage [7]. Consequently, it is crucial to develop effective strategies to alleviate the impact associated with DoS attacks and ensure the resilient operation of MASs. First, it is essential to develop a reasonable model for DoS attacks, as it plays an essential role in the related system analysis and synthesis. To establish a model that more accurately reflects the actual attack scenario, DoS attacks with limitations on duration and frequency were introduced for the first time in [8]. According to the basic model proposed above, the distributed DoS attack model was improved, allowing malicious adversaries to selectively disrupt individual channels or multiple channels concurrently, rather than interfering with all channels simultaneously [9]. The distributed estimation problem with DoS attacks was

2168-2216 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

examined [10], and sufficient criteria were derived to assure the stability of the estimation error system. The secure consensus problem for heterogeneous leader-follower MASs was addressed, with the system being transformed into a switching system according to different stages [11]. In the previously mentioned literature, the control input was typically set to zero during attack intervals, rendering the system open-loop. To reduce conservatism and fully utilize the system's data information, the precise calculation of control input during DoS attack intervals was considered, and a distributed event-based control strategy was developed accordingly [12]. Bipartite consensus control was investigated for MASs with antagonistic interactions and DoS attacks [13].

In addition, the issue of input saturation frequently arises, primarily because of the constraints of the physical actuator or the necessity for safety operations, such as the motor speed and power amplifiers. It is undeniable that input saturation could cause nonlinearity and impact the performance of agents [14]. Substantial progress has been made in addressing saturation-related issues [15], [16], [17]. The saturation avoidance method, which was known as low-gain feedback technique [18], was employed to solve the consensus problem. The core idea is selecting a minimal feedback gain to avoid input saturation, transforming systems with any initial states into linear ones without saturation nonlinearity. However, due to its conservative nature, the low-gain feedback could result in slower response time when handling input saturation which might impact the system's performance. Group consensus was investigated with input saturation based on sector-bounded conditions, which transformed the saturation nonlinearity into sector inequality [19]. Instead of avoiding saturation, the sector-bounded method processes saturation directly. Global non-negative consensus problem for a class of positive multiagents was studied, whose actuator was subject to sector-bounded saturation [20]. Asymmetric input saturation was considered and consensus was achieved for first-order MASs with directed topology [21].

Note that in the aforementioned works, continuous communication and control signals are required to design classical consensus control protocol, which necessitate abundant computation resources and ideal communication environment [22]. However, these requirements may not always be feasible in practical scenarios with limited bandwidth. Event-triggered mechanism (ETM) is designed to enable data exchange between agents only at specific instants triggered by a predefined event triggered mechanism. This event-triggered sampling strategy can effectively alleviate communication burden while maintaining desired performance of the controlled systems [23]. In recent years, the so-called dynamic triggering mechanisms (DETMs) were first carried out in [24] and then applied to MASs [25], [26], [27]. DETMs incorporate internal dynamic variables in the threshold conditions, distinguishing them from static ETMs (SETMs). The inclusion of internal dynamic variables enhances data transmission efficiency and reduces communication consumption by extending interevent intervals [24]. Two distributed dynamic event triggered principles were proposed to work on the consensus problem for single integrator agents [28]. A distributed DETM-based control protocol was designed and consensus was achieved for linear MASs with exponential convergence rate [29]. The time-varying output formation problem for heterogeneous MASs was investigated with node- and edge-based DETM controllers [30]. However, a notable shortcoming in the previous works [24], [28], [29], [30] is the necessity for continuous monitoring of neighboring agents' states, since the design of DETMs involves real-time states of the agent and its neighbors. More resource-efficient methods are desired to improved. To our understanding, there are few works investigating secure consensus for MASs with the combined impact of DoS attacks and input saturation while implementing DETMs to reduce the communication consumption and avoiding continuous monitoring. This serves as the motivation for this article.

Motivated by the aforementioned discussions, designing a resource-efficient control protocol to attain secure consensus for general MASs with input saturation and DoS attacks remains an open and challenging problem. To deal with these problems, a state-prediction-based DETM is proposed, aiming to alleviate over-consumption of communication resource and avoid continuous monitoring of neighbors' state information. Sector-bounded method is adopted to handle the input saturation. Efficient controller is given to guarantee secure consensus of MASs under unreliable environment. The correlation between consensus, DoS attacks, input saturation, and triggering times is disclosed. The main contributions are given as follows.

1) A unified framework is proposed for MASs that operate resource-efficiently in environments potentially subject to DoS attacks and input saturation. By employing the sector-bounded method and state-prediction-based DETM control protocol, consensus conditions of MASs can be derived by multi-Lyapunov function methods. Additionally, the occurrence of Zeno behavior is excluded by proving the existence of a positive lower bound on the interevent intervals.
2) Compared to [24], [28], [29], and [30], in which continuous monitoring of neighboring agents' states is needed, the DETM proposed in this article offers a more effectively solution. Based on state prediction, it allows the determination for triggering instants without the requirement of continuous real-time communication, thereby enhancing the operational efficiency of the systems.
3) A state-prediction-based DETM control protocol is proposed to address the secure consensus problem of MASs suffering from input saturation and DoS attacks, which brings in difficulties and challenges for convergence analysis and consensus conditions derivation. Compared ETM presented in [10] and [31], the decentralized DETM leveraging internal dynamic variable can ensure larger interevent intervals and efficiently transmit information. It can significantly relieve the burden of communication.

Notations: $\mathbb{N}$ is the natural number set. The real number set is expressed as $\mathbb{R}$. $\mathbb{R}_{\geq 0}$ represents the set of non-negative real number. $\mathbb{R}^m$ and $\mathbb{R}^{m\times m}$ indicate m-dimension vector and the $m \times m$ real matrix, respectively. For a certain

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

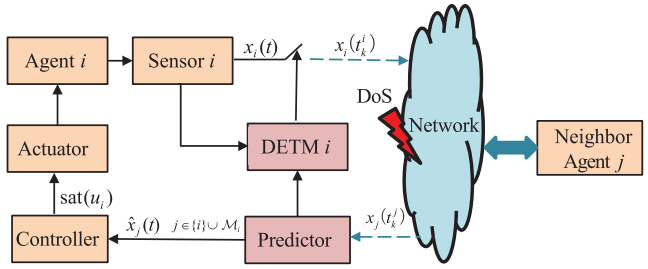


Fig. 1. Framework.

matrix P, $P > 0$, P^{T}, $\lambda(P)$ and $\|P\|$ denotes as positive matrix, corresponding transpose, eigenvalue and induced matrix 2-norm, respectively. For a certain vector $x_i \in \mathbb{R}^N$, the stack vector $[x_1^{\mathrm{T}}(t), x_2^{\mathrm{T}}(t), \ldots, x_N^{\mathrm{T}}(t)]^{\mathrm{T}}$ is expressed as $\mathrm{col}\{x_1(t), x_2(t), \ldots, x_N(t)\}$. $\mathbf{1_n} \in \mathbb{R}^{n\times 1}$ indicates that its entries are all 1. I_N indicates the N dimension identity matrix. For certain finite sets $\mathcal{S}_1$ and $\mathcal{S}_2$, $|\mathcal{S}_1|$ means the cardinality of $\mathcal{S}_1$, and $\mathcal{S}_1\backslash\mathcal{S}_2$ means relative complement of $\mathcal{S}_1$ in $\mathcal{S}_2$. The Kronecker product is denoted as the symbol $\otimes$, for given matrices $\mathcal{R} \in \mathbb{R}^{n\times m}$ and $\mathcal{H} \in \mathbb{R}^{p\times q}$, $\mathcal{R}\otimes\mathcal{H} \in \mathbb{R}^{np\times mq}$.

II. Problem Formulation and Preliminaries

A. Algebric Graph Theory

Considering an undirected graph topology $\mathcal{G} = \{\mathcal{M}, \mathcal{B}, \mathcal{A}\}$, $\mathcal{M} = \{m_1, m_2, \ldots, m_N\}$ indicates the agent set and the edge set is expressed as $\mathcal{B} \subseteq \mathcal{V}\times\mathcal{V}$. The adjacency matrix is expressed as $\mathcal{A} = [\pi_{ij}] \in \mathbb{R}^{N\times N}$. $\mathcal{G}$ is undirected when $(m_i, m_j) \in \mathcal{B}$ and $(m_j, m_i) \in \mathcal{B}$ satisfy simultaneously. If $(m_i, m_j) \in \mathcal{B}$, then the corresponding elements in $\mathcal{A}$ is $\pi_{ij} = a_{ji} = 1$, or $\pi_{ij} = a_{ji} = 0$ otherwise. Moreover, suppose $(m_i, m_i) \notin \mathcal{B}$ which means that the self-edge does not exist. The Laplacian matrix is represented as $\mathcal{L} = [\iota_{ij}] \in \mathbb{R}^{N\times N}$ with $\iota_{ij} = -\pi_{ij}$ and $\iota_{ii} = \sum_{j=1, j\neq i}^{N} \pi_{ij}$. $\mathcal{M}_i$ is the neighbor set of agent i, *i.e.*, $\mathcal{M}_i = \{j \in \mathcal{M} | (m_j, m_i) \in \mathcal{B}\}$. $\mathcal{G}$ is connected if there exists a path between arbitrary two agents.

B. Problem Formulation

The framework for MASs subject to malicious DoS attacks, input saturation, and state-prediction-based DETM consensus protocol is designed in Fig. 1.

First, a MAS contains N agents and is connected through an undirected topology $\mathcal{G}$, the state dynamics of agent i is given as follows:

$$\dot{x}_i(t) = Ax_i(t) + B\mathrm{sat}(u_i(t)) \tag{1}$$

where $x_i(t) \in \mathbb{R}^{n_x}$ and $\mathrm{sat}(u_i(t)) \in \mathbb{R}^{n_u}$ are the state and saturated input, respectively. For every element in vector u_i, we have $\mathrm{sat}(u_i) = [\mathrm{sat}(u_{i1}), \mathrm{sat}(u_{i2}), \ldots, \mathrm{sat}(u_{in_u})]^{\mathrm{T}}$, $\mathrm{sat}(u_{ip}) = \mathrm{sign}(u_{ip})\min\{\|u_{ip}\|, u_{\max}\}$, u_{ip} is the pth element of vector u_i. A, B and C are the system matrices of appropriate dimension.

To proceed with, we have the following fairly standard assumptions.

Assumption 1: The topology $\mathcal{G}$ is undirected and connected.

Assumption 2: The pair (A, B) is stabilizable.

The commonly used control protocol without saturated control input [32] for the MAS (1) is given as

$$u_i(t) = -K\sum_{j\in\mathcal{M}_i} \pi_{ij}\big(x_i(t) - x_j(t)\big) \tag{2}$$

to achieve consensus. Notice that the continuous communication between agent i and adjacent neighbors is needed to implement this kind control protocol. Such requirement is often impractical in physical application and causes the waste of resources seriously. To avert continuous communication, agent i only needs to transfer state information to adjacent neighbors at some necessary intermittent instants $\{t_k^i\}_{k\in\mathbb{N}}$, which are called triggering instants. Then, the DETM is introduced to alleviate communication burden and save energy, which will be designed later.

Lemma 1 [33]: Define $\Psi(u_i) = u_i - \mathrm{sat}(u_i)$, if the elements in $u_i \in \mathbb{R}^m$ and $\varpi_i \in \mathbb{R}^m$ belong to the set $\{u_i \in \mathbb{R}^m, \varpi_i \in \mathbb{R}^m, -u_{\max} \le u_{ip} - \varpi_{ip} \le u_{\max}\}$, then the saturation nonlinearity $\Psi(u_i)$ satisfies

$$\Psi^{\mathrm{T}}(u_i)\Upsilon(\Psi(u_i) + \varpi_i) \le 0$$

where $\Upsilon \in \mathbb{R}^{m\times m}$ denotes a diagonal positive matrix.

Sector-bounded method is adopted to deal with the saturation, Lemma 1 is the basis for utilizing the saturation-processing method. The system with saturation limition is difficult to analyze by using common Lyapunov stability theory, so the saturation term needs to be dealt with in advance. The sector-bounded condition provides an effective solution for transforming the stability analysis of the system into solvable LMIs.

C. Denial-of-Service Attack Model

As depicted in Fig. 1, agent i not only sends its data, but also receives data from its neighbors via communication networks. The data transmission process is susceptible to threats posed by malicious cyber-attacks. DoS attacks can result in communication jam without the requirement of prior knowledge of the controlled system, yet they can severely compromise the system performance and lead to considerable losses. Under the influence of DoS attacks, if DoS attacks remain perpetually active, rendering that the exchange of state information is impossible. Consequently, the agent cannot update its own behavior based on the status of adjacent neighbor agents, nor can it broadcast its state to other agents.

Now, we consider a more realistic form of aperiodic DoS attacks whose energy is limited. The attackers can launch the attack in an active period and also have to enter a sleeping period to store enough energy for another attack. So whole time could be divided into two stages, one is a safe interval that the agents are well-communicated and the system is operating normally, another is an attacked interval that communication is prohibited and the system maybe fail.

Let $\{d_n\}_{n\in\mathbb{N}}$ be the sequence of DoS *off/on* transition instants, which denote that the state of attacks shifts from dormant to active. Define nth DoS attack interval as $D_n = [d_n, d_n+\tau_n)$, where $\tau_n \ge 0$ represents the duration of nth attack interval, the communication during this period is interrupted. Notice that if $\tau_n = 0$, then D_n becomes single pulse attack.

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Let $\mathcal{D}(t_0, t)$ represent the union set of attack intervals, $\mathcal{S}(t_0, t)$ denotes the union set of safe intervals.

$$\mathcal{D}(t_0, t) = \bigcup_{n\in\mathbb{N}} D_n \bigcap [t_0, t]$$
$$\mathcal{S}(t_0, t) = [t_0, t] \backslash \mathcal{D}(t_0, t).$$

Obviously, $\mathcal{D}(t_0, t) \bigcup \mathcal{S}(t_0, t) = t - t_0$.

Based on [8], let $\mathcal{F}(t_0, t)$ represent the frequency of DoS *off/on* transitions during the period (t_0, t).

Assumption 3 ([8], Frequency): For $0 \le t_0 \le t$, there exist $F_1 \in \mathbb{R}_{\ge 0}$, $F_2 \in \mathbb{R}_{>0}$ such that

$$\mathcal{F}(t_0, t) \le F_1 + \frac{t - t_0}{F_2}. \tag{3}$$

Assumption 4 ([8], Duration): For $0 \le t_0 \le t$, there exist $T_1 \in \mathbb{R}_{\ge 0}$, $T_2 \in \mathbb{R}_{>1}$ such that

$$|\mathcal{D}(t_0, t)| \le T_1 + \frac{t - t_0}{T_2}. \tag{4}$$

Compared to other existing models of DoS attacks which are periodic [34], [35], [36] or follow some probability distribution, the Assumptions 3 and 4 only make hypothesis on frequency and duration of DoS attacks. This kind of hypothesis is more general and aligned with realistic attack strategy.

Remark 1: The hypothesis is rational that there usually exist energy constraints of malicious adversaries or some approaches can mitigate DoS attacks, such as high-pass filters. The energy consumption of attacks is proportional to the duration of the active attacks and the number of *off/on* transition moments, which can be modeled into an intermittent pattern as Assumptions 3 and 4. It's necessary to model DoS attacks in terms of both duration and frequency. If DoS attacks occur frequently or last for a very long time, the data cannot be received and transmitted to neighbors, which implies that all the information exchange attempts keep failing and consensus for MASs cannot be realized intuitively. The Assumptions 3 and 4 are fairly mild and common, which have been considered in [10], [12], and [37]. The parameter F_2 can be interpreted as the average dwell time between successive DoS *off/on* transitions, F_1 is the regularization term. Similarly, Assumption 4 reflects the slow-on-average property, the total duration of DoS attacks would not exceed a certain fraction of time, *i.e.*, $1/T_2$, and T_1 serves as the regularization term. If the values of F_1 and T_1 are too large, indicating that the DoS attacks can persist for a prolonged period at the beginning, then the system may fail to fulfill the desired performance within an acceptable timeframe, which can result in degraded performance.

D. Predictor-Based Event-Triggered Control Protocol

To further aviod continuous monitoring and communication, each agent is equipped with a predictor and DETM. The predictor can estimate the state information of itself and its neighbors between adjacent triggering instants by utilizing the latest received information. The predictor embedded in agent i is given as

$$\hat{x}_j(t) = e^{A\left(t - t_k^j\right)} x_j\left(t_k^j\right), t \in \left[t_k^j, t_{k+1}^j\right) \tag{5}$$

where $\hat{x}_j(t)$ is the prediction of agent i and adjacent neighbors, for $j \in \{i\} \cup \mathcal{M}_i$. As the system framework depicted in Fig. 1, the states of agent i are transmitted only when the condition of DETM is satisfied, if there are no predictions on the values of itself, then the controller does not receive any data between two adjacent triggering moments. By using the predicted state values between two adjacent triggering instants, the system's responsiveness to dynamic changes is improved.

For agent i, current triggering instant is denoted as t_k^i, then the next triggering instant t_{k+1}^i is determined by the following DETM:

$$\begin{cases} t_{k+1}^i = \inf\left\{t > t_k^i | f_i\left(e_i(t), \eta_i(t), \cup_{j\in\mathcal{M}_i} \hat{x}_j(t)\right) > 0\right\} \\ \dot{\eta}_i(t) = g_i\left(e_i(t), \eta_i(t), \cup_{j\in\mathcal{M}_i} \hat{x}_j(t)\right). \end{cases} \tag{6}$$

The function $f_i(\cdot)$ is formulated as

$$f_i(t) = -\theta_i \eta_i(t) - \alpha \left\| \sum_{j\in\mathcal{M}_i} \pi_{ij}\left(\hat{x}_i(t) - \hat{x}_j(t)\right) \right\|^2 + e_i^{\mathrm{T}}(t) \Omega e_i(t) \tag{7}$$

where $e_i(t) = \hat{x}_i(t) - x_i(t)$ is the estimation error among the predicted value and actual value of agent i. α, θ_i are positive constant parameters, and the positive-definite matrix Ω is to be determined later.

The dynamics of internal state $\eta_i(t)$ satisfies

$$\dot{\eta}_i(t) = -\beta_i \eta_i(t) + \alpha \left\| \sum_{j\in\mathcal{M}_i} \pi_{ij}\left(\hat{x}_i(t) - \hat{x}_j(t)\right) \right\|^2 - e_i^{\mathrm{T}}(t) \Omega e_i(t) \tag{8}$$

where $\beta_i > 0$ and the initial internal variable $\eta_i(0) > 0$. Assume that when $t \in D_n$, $\dot{\eta}_i(t) = 0$.

The control input based on predicted value and DETM is given

$$u_i(t) = -K \sum_{j\in\mathcal{M}_i} \pi_{ij}\left(\hat{x}_i(t) - \hat{x}_j(t)\right). \tag{9}$$

Remark 2: The DETM (6) involves an internal dynamic variable η_i which is always equal or greater than zero, making the interevent intervals greater than SETM. Thus, the communication times between agents can be reduced effectively based on DETM, thus saving energy. Notice that when the parameter $\theta_i = 0$, the DETM (6) degenerates into SETM counterpart as follows:

$$t_{k+1}^i = \inf\left\{ t > t_k^i | e_i(t)^{\mathrm{T}} \Omega e_i(t) > \alpha \left\| \sum_{j\in\mathcal{M}_i} \pi_{ij}\left(\hat{x}_i(t) - \hat{x}_j(t)\right) \right\|^2 \right\}. \tag{10}$$

Therefore, the SETM (10) is a special case of DETM (6). The following corollary illustrates that the next triggered time is larger determined by DETM than that given by SETM for a given state.

Corollary 1: Let $k \in \mathbb{N}$, $x(t_n^i) \in \mathbb{R}^n$ and $\eta_i(t_n^i) \ge 0$, $t_{d,n+1}^i$ is the next triggering instant decided by DETM (6), and $t_{s,n+1}^i$ denotes the next triggering instant decided by SETM (10), then $t_{s,n+1}^i < t_{d,n+1}^i$.

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Proof: The proof is completed by contradiction. First, suppose that $t^i_{s,n+1}$ satisfies the condition of SETM (10), and $t^i_{s,n+1} > t^i_{d,n+1}$. This implies $t^i_{d,n+1}$ does not meet (10), accordingly, we have that

$$\alpha\left\|\sum_{j\in\mathcal{M}_i}\pi_{ij}\big(\hat{x}_i\big(t^i_{d,n+1}\big)-\hat{x}_j\big(t^i_{d,n+1}\big)\big)\right\|^2 > e_i\Big(t^{i^-}_{d,n+1}\Big)^{\mathrm{T}}\Omega e_i\Big(t^{i^-}_{d,n+1}\Big). \tag{11}$$

Since $t^i_{d,n+1}$ is the triggering instant determined by DETM (6), we have

$$\begin{aligned}0 &> \theta_i\eta_i\big(t^i_{d,n+1}\big)+\alpha\left\|\sum_{j\in\mathcal{M}_i}\pi_{ij}\big(\hat{x}_i\big(t^i_{d,n+1}\big)-\hat{x}_j\big(t^i_{d,n+1}\big)\big)\right\|^2\\ &\quad - e_i\Big(t^{i^-}_{d,n+1}\Big)^{\mathrm{T}}\Omega e_i\Big(t^{i^-}_{d,n+1}\Big)\\ &> \alpha\left\|\sum_{j\in\mathcal{M}_i}\pi_{ij}\big(\hat{x}_i\big(t^i_{d,n+1}\big)-\hat{x}_j\big(t^i_{d,n+1}\big)\big)\right\|^2\\ &\quad - e_i\Big(t^{i^-}_{d,n+1}\Big)^{\mathrm{T}}\Omega e_i\Big(t^{i^-}_{d,n+1}\Big).\end{aligned}$$

This contradicts (11), hence $t^i_{s,n+1} < t^i_{d,n+1}$. Thus, this completes the proof. ∎

Remark 3: Supposed that the DETM-based control strategy is effectively executed through TCP-like communication networks, the acknowledgement-based mechanism plays a crucial role in ensuring efficient state information transmission in practical implementation. The acknowledgement process can be described as follows. Upon successful reception of the transmitted state information by the controller, an acknowledgement signal would be promptly sent to the DETM side, thereby confirming the successful completion of data transmission. Subsequently, the DETM, in response to receiving the acknowledgement signal, initiates the transmission of a validation signal back to the controller. If the controller does not receive the validation signal in time, then it recognizes that it is currently operating under DoS attacks. In a situation where the DETM fails to receive the acknowledgement signal, it becomes aware that the current moment is within the attack interval. In this scenario, the condition (6) may be violated since the data updating is unsuccessful. The DETM module will stop calculations to save energy, but will keep transmitting the latest successful data until it receives the acknowledge signal to confirm the recovery of communication, which implies the end of DoS attacks. Once the transmission is recovered, the DETM can receive the data information to update the condition and operates normally in safe intervals.

E. Control Objective

The goal of this article is to design a secure consensus control protocol for each agent with state-prediction-based DETM, so that the local consensus of agents can be reached despite the existence of DoS attacks and input saturation, if for any initial state $x_i(0)$ belongs to a specific set $\psi(P,\varepsilon)$. Therefore there two main task: 1) the utilization of communication resource can be effectively reduced and continuous monitoring can be avoided based on such a state-prediction-based DETM transmission strategy and 2) the states of all agents can attain a common state, such that

$$\lim_{t\to+\infty}\|x_i(t)-x_j(t)\|=0 \tag{12}$$

holds though communication network is unreliable and limited.

III. Main Result

A novel consensus criterion for MAS (1) under aperiodic DoS attacks and input saturation is presented. And the sufficient condition is given to eliminate Zeno behavior.

For simplicity of notation, the following notations are introduced:

$$\begin{aligned}x(t)&=\mathrm{col}_N\{x_i(t)\},\quad y(t)=\mathrm{col}_N\{y_i(t)\}\\ u(t)&=\mathrm{col}_N\{u_i(t)\},\quad e(t)=\mathrm{col}_N\{e_i(t)\}\\ \hat{x}(t)&=\mathrm{col}_N\{\hat{x}_i(t)\}.\end{aligned} \tag{13}$$

The average state of MAS is given as

$$\tilde{x}(t)\triangleq\frac{1}{N}\sum_{i=1}^{N}x_i(t)\triangleq\frac{1}{N}\big(\mathbf{1}_N{}^{\mathrm{T}}\otimes I_{n_x}\big)x(t). \tag{14}$$

The disagreement variable is defined as

$$\bar{x}(t)\triangleq x(t)-\big(\mathbf{1}_N\otimes I_{n_x}\big)\tilde{x}(t)=\big(\mathcal{L}_N\otimes I_{n_x}\big)x(t) \tag{15}$$

where $\bar{x}(t)$ means the error between the states of each agent and $\tilde{x}(t)$, $\mathcal{L}_N=I_N-\frac{1}{N}\mathbf{1}_N\mathbf{1}_N^{\mathrm{T}}$. The properties of $\mathcal{L}$ and $\mathcal{L}_N$ are given as follows:

$$\mathcal{L}\mathcal{L}_N=\mathcal{L}_N\mathcal{L}=\mathcal{L},\ \ \mathcal{L}_N\mathcal{L}_N=\mathcal{L}_N. \tag{16}$$

Then combined with saturation control input, state-prediction-based control input (9) and MAS (1), the dynamic of disagreement variable is given

$$\begin{aligned}\dot{\bar{x}}(t)&=(I_N\otimes A)\bar{x}(t)+(\mathcal{L}_N\otimes B)\mathrm{sat}(-(\mathcal{L}\otimes K)(\bar{x}+e(t)))\\ &=(I_N\otimes A-\mathcal{L}\otimes BK)\bar{x}(t)-(\mathcal{L}\otimes BK)e(t)\\ &\quad-(\mathcal{L}_N\otimes B)\Psi(u)\end{aligned} \tag{17}$$

where $u=-(\mathcal{L}\otimes K)(\bar{x}(t)+e(t))$.

Now, let's define the specific set as $\psi(P,\varepsilon)\triangleq\{x_i(t)|\sum_{i=1}^{N}\bar{x}_i^{\mathrm{T}}P\bar{x}_i\le\varepsilon),\bar{x}_i=x_i(t)-\tilde{x}(t)\}$. ε is a positive scalar and P is a positive defined matrix.

Remark 4: Notice that the matrix $\mathcal{L}_N$ can be treated as Laplacian matrix of N agents via a completed graph of which all the edges have a weight of $(1/N)$. A completed graph means that every pair of distinct vertices is connected by a unique edge. According to the property of connected Laplacian matrix, $\mathcal{L}_N$ only has a simple zero eigenvalue with its corresponding eigenvector $\mathbf{1}_N$. If $\bar{x}(t)=0$, then $x_1(t)=x_2(t)=\cdots=x_N(t)$ holds, which means that if the disagreement system (17) can achieve asymptotic stable, naturally the consensus problem of MAS (1) is settled. Thus, the consensus problem of MAS (1) has been turned into the stability analysis problem of disagreement system (17).

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Lemma 2: For a connected and undirected graph $\mathcal{G}$, there exists a unitary matrix $\Phi = [(\mathbf{1}_N/\sqrt{N}), \phi_{2:N}]$ such that $\Phi^{\mathrm{T}}\mathcal{L}\Phi = \tilde{\mathcal{J}} = \mathrm{diag}\{0, \sigma_2, \sigma_3, \ldots, \sigma_N\} = \mathrm{diag}\{0, \mathcal{J}\}$, σ_i is the eigenvalue of $\mathcal{L}$, $\Phi\Phi^{\mathrm{T}} = \Phi^{\mathrm{T}}\Phi = I_N$.

Theorem 1: Consider a MAS (1) subject to malicious DoS attacks and input saturation, and it satisfies with Assumption 1 and 2. For some given parameters $\theta_i > 0$, $\beta_i > 0$, $\gamma_1 > 0$, $c_2 > 0$, $\alpha > 0$, secure consensus can be achieved with the DETM (6) and state-prediction-based controller (9), if DoS attacks meet the following:

$$\frac{\ln(\lambda_1\lambda_2)}{F_2} + \frac{c_1 + c_2}{T_2} < c_1, \tag{18}$$

and if there exist a matrix $Q = Q^{\mathrm{T}} > 0$ satisfying $A^{\mathrm{T}}Q + QA - c_2Q < 0$ and positive-definite matrices P, Ω, and Υ, matrix $\mathscr{W}$ such that the following inequalities hold:

$$\begin{bmatrix} \tilde{\Sigma}_{11} & 0 & \bar{\Sigma}_{13} & 0 & \tilde{\Sigma}_{15} & \tilde{\Sigma}_{16} \\ * & -2\Upsilon & 0 & 0 & 0 & 0 \\ * & * & \bar{\Sigma}_{33} & 0 & \bar{\Sigma}_{35} & 0 \\ * & * & * & -\Omega & 0 & 0 \\ * & * & * & * & \tilde{\Sigma}_{55} & \tilde{\Sigma}_{56} \\ * & * & * & * & * & \tilde{\Sigma}_{66} \end{bmatrix} < 0 \tag{19}$$

$$\begin{bmatrix} I_N \otimes P & I_N \otimes \mathscr{W}^{\mathrm{T}} \\ * & \varepsilon^{-1}u_{\max}^2 \end{bmatrix} \geq 0 \tag{20}$$

where $c_1 = \min_i\{\gamma_1, \beta_i\}$, $\lambda_1 = \max\{1, (\lambda_{\max}(Q)/\lambda_{\min}(P))\}$, $\lambda_2 = \max\{1, (\lambda_{\max}(P)/\lambda_{\min}(Q))\}$, $\tilde{\Sigma}_{11} = I_{N-1}\otimes(PA+A^{\mathrm{T}}P) - 2\mathcal{J}\otimes PBK + \gamma_1(I_{N-1}\otimes P)$, $\bar{\Sigma}_{13} = -\mathcal{J}_N\otimes PB + \mathcal{J}\otimes K^{\mathrm{T}}\Upsilon^{\mathrm{T}} - I_{N-1}\otimes\mathscr{W}^{\mathrm{T}}\Upsilon^{\mathrm{T}}$, $\tilde{\Sigma}_{15} = -\mathcal{J}\otimes PBK$, $\tilde{\Sigma}_{16} = \mathcal{J}\otimes I_{n_x}$, $\bar{\Sigma}_{33} = -2(I_{N-1}\otimes\Upsilon)$, $\bar{\Sigma}_{35} = \mathcal{J}\otimes\Upsilon K$, $\tilde{\Sigma}_{55} = -I_{N-1}\otimes\Omega$, $\tilde{\Sigma}_{56} = \mathcal{J}\otimes I_{n_x}$, $\tilde{\Sigma}_{66} = -(1/\alpha)I_{(N-1)n_x}$.

Moreover, Zeno behavior can be eliminated, if the following equation holds:

$$\begin{bmatrix} I_N\otimes\Omega & \tilde{\mathcal{J}}\otimes I_{n_x} \\ * & (1/\alpha(1+a))I_{Nn_x} \end{bmatrix} > 0 \tag{21}$$

where $a > 0$.

Proof: For clarity, the proof can be separated into two parts: consensus analysis and the exclusion of Zeno behavior.

Part 1: Consensus analysis. Since the time axis is divided into two different modes depending on whether the current moment is under DoS attacks or not, the discussion can be divided into two cases.

Case 1: During the safe intervals $t \in S_{n-1} = [d_{n-1} + \tau_{n-1}, d_n)$, first notice that in terms of DETM (6) with the function (7) and dynamic of internal state $\eta_i(t)$ (8), we obtain

$$\dot{\eta}_i(t) \geq -(\beta_i + \theta_i)\eta(t). \tag{22}$$

By the comparison of differential inequality and the initial state $\eta_i(0) > 0$, thus we have

$$\eta_i(t) \geq e^{-(\beta_i+\theta_i)t}\eta_i(0) > 0. \tag{23}$$

Then the following Lyapunov function is chosen:

$$W_1(\bar{x}(t), \eta_i(t)) = V(t) + \sum_{i=1}^{N}\eta_i(t) \tag{24}$$

where $V(t) = \bar{x}^{\mathrm{T}}(t)(I_N\otimes P)\bar{x}(t)$.

Let $\varpi_i = u_i + \mathscr{W}\bar{x}_i(t)$, where $\mathscr{W}$ is a matrix with compatible dimension. If $|\mathscr{W}\bar{x}_i(t)| \leq u_{\max}$, then based on Lemma 1, we have

$$\Psi^{\mathrm{T}}(u)(I_N\otimes\Upsilon)\big(\Psi(u) + u + \bar{\mathscr{W}}\bar{x}(t)\big) \leq 0 \tag{25}$$

where $\bar{\mathscr{W}} = I_N\otimes\mathscr{W}$. Applying Schur's complement to the inequality (20), we can obtain that $\varepsilon(I_N\otimes\mathscr{W})^{\mathrm{T}}(I_N\otimes\mathscr{W}) \leq u_{\max}^2(I_N\otimes P)$, then we have $\sum_{i=1}^{N}(\mathscr{W}\bar{x}_i(t))^{\mathrm{T}}(\mathscr{W}\bar{x}_i(t)) \leq \varepsilon^{-1}u_{\max}^2\sum_{i=1}^{N}\bar{x}_i^{\mathrm{T}}(t)P\bar{x}_i(t)$. For $\bar{x}_i(t) \in \psi(P,\varepsilon)$, we can get that $|\mathscr{W}\bar{x}_i(t)|^{\mathrm{T}}|\mathscr{W}\bar{x}_i(t)| < \sum_{i=1}^{N}(\mathscr{W}\bar{x}_i(t))^{\mathrm{T}}(\mathscr{W}\bar{x}_i(t)) \leq \varepsilon^{-1}u_{\max}^2\sum_{i=1}^{N}\bar{x}_i^{\mathrm{T}}(t)P\bar{x}_i(t) \leq u_{\max}^2$. The ellipsoid $\psi(P,\varepsilon)$ is included in the domain $|\mathscr{W}\bar{x}| \leq u_{\max}$.

Combining the disagreement system (17) and DETM function (7), then the derivative of $W_1(t)$ is given by

$$\begin{aligned}
\dot{W}_1 &= \bar{x}^{\mathrm{T}}(t)\big[I_N\otimes\big(PA + A^{\mathrm{T}}P\big) - 2\mathcal{L}\otimes PBK\big]\bar{x}(t) \\
&\quad -2\bar{x}^{\mathrm{T}}(t)(\mathcal{L}\otimes PBK)e(t) - 2\bar{x}^{\mathrm{T}}(t)(\mathcal{L}_N\otimes PB)\Psi(u) \\
&\quad +\sum_{i=1}^{N}\left(-\beta_i\eta_i(t) + \alpha\Big\|\sum_{j\in\mathcal{M}_i}(\hat{x}_i(t) - \hat{x}_j(t))\Big\|^2 - e_i^{\mathrm{T}}(t)\Omega e_i(t)\right) \\
&\leq \bar{x}^{\mathrm{T}}(t)\big[I_N\otimes\big(PA + A^{\mathrm{T}}P\big) - 2\mathcal{L}\otimes PBK\big]\bar{x}(t) \\
&\quad -2\bar{x}^{\mathrm{T}}(t)(\mathcal{L}\otimes PBK)e(t) - 2\bar{x}^{\mathrm{T}}(t)(\mathcal{L}_N\otimes PB)\Psi(u) \\
&\quad -e^{\mathrm{T}}(t)(I_N\otimes\Omega)e(t) + \alpha\big\|\big(\mathcal{L}\otimes I_{n_x}\big)(e(t) + \bar{x}(t))\big\|^2 \\
&\quad -2\Psi^{\mathrm{T}}(u)(I_N\otimes\Upsilon)\big(\Psi(u) + u + \bar{\mathscr{W}}\bar{x}(t)\big) - \sum_{i=1}^{N}\beta_i\eta_i(t) \\
&\leq -\gamma_1\bar{x}^{\mathrm{T}}(t)(I_N\otimes P)\bar{x}(t) - \sum_{i+1}^{N}\beta_i\eta_i(t) + \varsigma^{\mathrm{T}}\Sigma\varsigma
\end{aligned} \tag{26}$$

where $\varsigma = [\bar{x}^{\mathrm{T}}(t), \Psi^{\mathrm{T}}(u), e^{\mathrm{T}}(t)]^{\mathrm{T}}$ and

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} \\ * & \Sigma_{22} & \Sigma_{23} \\ * & * & \Sigma_{33} \end{bmatrix} \tag{27}$$

with $\Sigma_{11} = I_N\otimes(PA + A^{\mathrm{T}}P) - 2\mathcal{L}\otimes PBK + \gamma_1(I_N\otimes P) + \alpha(\mathcal{L}^2\otimes I_{n_x})$, $\Sigma_{12} = -\mathcal{L}_N\otimes PB + \mathcal{L}\otimes K^{\mathrm{T}}\Upsilon^{\mathrm{T}} - I_N\otimes\mathscr{W}^{\mathrm{T}}\Upsilon^{\mathrm{T}}$, $\Sigma_{13} = -\mathcal{L}\otimes PBK + \alpha(\mathcal{L}^2\otimes I_{n_x})$, $\Sigma_{22} = -2I_N\otimes\Upsilon$, $\Sigma_{23} = \mathcal{L}\otimes\Upsilon K$, $\Sigma_{33} = -I_N\otimes\Omega + \alpha(\mathcal{L}^2\otimes I_{n_x})$.

Denoting $\bar{\zeta} = \mathrm{diag}\{\Phi^{\mathrm{T}}\otimes I_{n_x}, \Phi^{\mathrm{T}}\otimes I_{n_u}, \Phi^{\mathrm{T}}\otimes I_{n_x}\}\zeta$, $\Phi^{\mathrm{T}}\mathcal{L}\Phi = \tilde{\mathcal{J}} = \mathrm{diag}\{0, \mathcal{J}\}$,$\Phi^{\mathrm{T}}\mathcal{L}_N\Phi = \tilde{\mathcal{J}}_N = \mathrm{diag}\{0, \mathcal{J}_N\}$. Since $(\Phi^{\mathrm{T}}\otimes I_{n_x})\bar{x}(t) = [\bar{x}_{\phi 1}^{\mathrm{T}}(t), \bar{x}_{\phi 2:N}^{\mathrm{T}}(t)]^{\mathrm{T}}$, $\bar{x}_{\phi 1}(t) = (\mathbf{1}/\sqrt{N})\sum_{i=1}^{N}\bar{x}(t) = 0$, thus we have $\zeta^{\mathrm{T}}\Sigma\zeta = \bar{\zeta}_{2:3N}^{\mathrm{T}}\bar{\Sigma}\bar{\zeta}_{2:3N}$, where

$$\bar{\Sigma} = \begin{bmatrix} \bar{\Sigma}_{11} & 0 & \bar{\Sigma}_{13} & 0 & \bar{\Sigma}_{15} \\ * & -2\Upsilon & 0 & 0 & 0 \\ * & * & \bar{\Sigma}_{33} & 0 & \bar{\Sigma}_{35} \\ * & * & * & -\Omega & 0 \\ * & * & * & * & \bar{\Sigma}_{55} \end{bmatrix} \tag{28}$$

with $\bar{\Sigma}_{11} = I_{N-1}\otimes(PA + A^{\mathrm{T}}P) - 2\mathcal{J}\otimes PBK + \gamma_1(I_{N-1}\otimes P) + \alpha(\mathcal{J}^2\otimes I_{n_x})$, $\bar{\Sigma}_{13} = -\mathcal{J}_N\otimes PB + \mathcal{J}\otimes K^{\mathrm{T}}\Upsilon^{\mathrm{T}} - I_{N-1}\otimes\mathscr{W}^{\mathrm{T}}\Upsilon^{\mathrm{T}}$, $\bar{\Sigma}_{15} = -\mathcal{J}\otimes PBK + \alpha(\mathcal{J}^2\otimes I_{n_x})$, $\bar{\Sigma}_{33} = -2(I_{N-1}\otimes\Upsilon)$, $\bar{\Sigma}_{35} = \mathcal{J}\otimes\Upsilon K$, $\bar{\Sigma}_{55} = -I_{N-1}\otimes\Omega + \alpha(\mathcal{J}^2\otimes I_{n_x})$.

Once the condition (19) is satisfied, then $\bar{\Sigma} < 0$. For $\gamma_1 > 0$, $\beta_i > 0$, $P > 0$, The derivative of Lyapunov function $W_1(t)$ can be obtained as

$$\dot{W}_1(t) \leq -c_1W_1(t). \tag{29}$$

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Denote $c_1 = \min_i\{\gamma_1, \beta_i\}$, by the comparison of differential inequality, we have that

$$W_1(t) \le e^{-c_1 t} W_1(0). \tag{30}$$

Consider that $W_1(t) > 0$ and $\dot{W}_1(t) < 0$, the asymptotically stable of disagreement system (17) is achieved. This means that consensus of MAS (1) can be reached during the safe intervals.

Case 2: During the attack intervals $t \in D_n = [d_n, d_n + \tau_n)$, notice that the communication between agents is jammed by the malicious attackers, the transmitted information cannot be propagated to its neighbors to update the DTEM, thus the DETM (6) may be not satisfied in this period. The DETM communication strategy works only during the period which communication is allowed, assume that the DETM doesn't work, *i.e.,* $\dot{\eta}_i(t) = 0$ during the DoS attack intervals. For the sake of energy conservation, the control input $u_i(t)$ is set as zero when the communication between agents is cut off.

And the Lyapunov candidate is considered as

$$W_2(t) = \bar{x}^{\mathrm{T}}(t)(I_N \otimes Q)\bar{x}(t) \tag{31}$$

then the derivation trajectory of $W_2(t)$ is calculated

$$\begin{aligned} \dot{W}_2 &= \bar{x}^{\mathrm{T}}(t)\big[I_N \otimes \big(QA + A^{\mathrm{T}}Q\big)\big]\bar{x}(t) \\ &< c_2 W_2(t) \end{aligned} \tag{32}$$

where $c_2 > 0$ satisfies $A^{\mathrm{T}}Q + QA - c_2 Q < 0$.

Case 3: Stability analysis of all horizon of time based on the above processes. Combined the inequalities (30) and (32), first we can get the two switching Lyapunov functions depending on whether current instant belongs to secure intervals S_{n-1} or attack intervals D_n, $n \in \mathbb{N}$.

$$W(t) = \begin{cases} e^{-c_1\left(t - d_{n-1} - \tau_{n-1}\right)} W_1(d_{n-1} + \tau_{n-1}), t \in S_{n-1} \\ e^{c_2(t-d_n)} W_2(d_n), t \in D_n. \end{cases} \tag{33}$$

Define two regulation parameters $\lambda_1 = \max\{1, (\lambda_{\max}(Q)/\lambda_{\min}(P))\}$, $\lambda_2 = \max\{1, (\lambda_{\max}(P)/\lambda_{\min}(Q))\}$. Therefore we have $Q < \lambda_1 P$, $P < \lambda_2 Q$. Thus, we have

$$\begin{cases} W_1\big((d_{n-1} + \tau_{n-1})^+\big) < \lambda_2 W_2\big((d_{n+1} + \tau_{n-1})^-\big), t \in S_{n-1} \\ W_2\big(d_n^+\big) < \lambda_1 W_1\big(d_n^-\big), t \in D_n. \end{cases}$$

Without a loss in generality, suppose that it is in a safe interval, $t \in S_{n-1}$, we have

$$\begin{aligned} W(t) &\le e^{-c_1\left(t - d_{n-1} - \tau_{n-1}\right)} W_1(d_{n-1} + \tau_{n-1}) \\ &\le \lambda_2 e^{-c_1\left(t - d_{n-1} - \tau_{n-1}\right)} e^{c_2 \tau_{n-1}} W_2(d_{n-1}) \\ &\le \lambda_1\lambda_2 e^{-c_1\left(t - d_{n-1} - \tau_{n-1}\right)} e^{c_2\tau_{n-1}} \\ &\qquad e^{-c_1\left(d_{n-1} - d_{n-2} - \tau_{n-2}\right)} W_1(d_{n-2} + \tau_{n-2}) \\ &\le \cdots \\ &\le (\lambda_1\lambda_2)^n e^{-c_1|\mathcal{S}(0,t)|} e^{c_2|\mathcal{D}(0,t)|} W_1(0). \end{aligned} \tag{34}$$

Note that $|\mathcal{S}(0,t)| = t - |\mathcal{D}(0,t)|$, substituting the constrains in Assumptions 3 and 4 into (34), then we have

$$\begin{aligned} W(t) &\le (\lambda_1\lambda_2)^{F_1 + \frac{t}{F_2}} e^{-c_1 t} e^{(c_1+c_2)\left(T_1 + \frac{t}{T_2}\right)} W_1(0) \\ &\le e^{\ln(\lambda_1\lambda_2)\left(F_1 + \frac{t}{F_2}\right)} e^{-c_1 t} e^{(c_1+c_2)\left(T_1 + \frac{t}{T_2}\right)} W_1(0) \\ &\le e^{\ln(\lambda_1\lambda_2)F_1 + (c_1+c_2)T_1} e^{\left(\frac{\ln(\lambda_1\lambda_2)}{F_2} + \frac{c_1+c_2}{T_2} - c_1\right)t} W_1(0) \\ &\le \rho e^{-\mu t} W_1(0) \end{aligned}$$

where $\rho = e^{\ln(\lambda_1\lambda_2)F_1 + (c_1+c_2)T_1}$, $\mu = c_1 - ([\ln(\lambda_1\lambda_2)/F_2] + [c_1 + c_2/T_2])$. Based on the condition (18), it's easy to know that $\mu > 0$, that is to say if DoS attacks satisfy the constraint (18). Then the disagreement system with the proposed controller is exponentially stable, the secure consensus of MAS (1) is exponentially achieved.

Part 2: Rule out Zeno behavior. This part proof is completed by contradiction. We hypothesise that Zeno phenomenon exhibits for any agent i. Then assume that there is a positive constant T_0 such that $\lim_{k\to\infty} t_k^i = T_0$. By the property and definition of limit, for arbitrary $\epsilon > 0$, there is an positive integer $N(\epsilon)$ such that $t_k^i \in (T_0 - \epsilon, T_0 + \epsilon)$, for $\forall k \ge N(\epsilon)$. Obviously, it's easy to obtain that $t_{N(\epsilon)+1}^i - t_{N(\epsilon)}^i < 2\epsilon$. According to the derivation of stability analysis above, the Lyapunov candidate $W(t)$ is bounded, *i.e.,* $W(t) < W_0$, where W_0 is the upper bound of $W(t)$. Accordingly, both the internal variable $\eta_i(t)$ and deviation state $\bar{x}_i$ are also bounded. Thus, there are positive constants $C_{\bar{x}}$ and C_η such that

$$\|\bar{x}_i\| < C_{\bar{x}}, \quad \|\eta_i(t)\| < C_\eta. \tag{35}$$

Based on the DETM function (7), for $t \in [t_k^i, t_{k+1}^i)$, using the stack form of the variables, we have

$$\begin{aligned} & e^{\mathrm{T}}(t)(I_N \otimes \Omega)e(t) \\ &\le \alpha\big\|\big(\mathcal{L} \otimes I_{n_x}\big)(\bar{x}(t) + e(t))\big\|^2 + \sum_{i=1}^{N} \theta_i \eta_i(t) \\ &\le \alpha(1+a)e^{\mathrm{T}}(t)\Big(\mathcal{L}^2 \otimes I_{n_x}\Big)e(t) + \sum_{i=1}^{N} \theta_i \eta_i(t) \\ &\quad + \alpha\left(1 + \frac{1}{a}\right)\bar{x}^{\mathrm{T}}\Big(\mathcal{L}^2 \otimes I_{n_x}\Big)\bar{x}(t). \end{aligned}$$

The second inequality holds by employing the Young's inequality, $2xy \le ax^2 + (1/a)y^2$. Thus we have

$$\begin{aligned} \|e(t)\|^2 \le \frac{1}{\lambda_{\min}(\Phi)}\Bigg[& \alpha\left(1 + \frac{1}{a}\right)\bar{x}(t)^{\mathrm{T}}\Big(\mathcal{L}^2 \otimes I_{n_x}\Big)\bar{x}(t) \\ & + \sum_{i=1}^{N} \theta_i \eta_i(t)\Bigg] \end{aligned} \tag{36}$$

where $\Phi = I_N \otimes \Omega - \alpha(1+a)(\mathcal{L}^2 \otimes I_{n_x}) > 0$, which is guaranteed by (21). Then it follows that:

$$\begin{aligned} \|e_i(t)\| &\le \sqrt{\frac{1}{\lambda_{\min}(\Phi)}\left[\alpha\left(1 + \frac{1}{a}\right)\|\mathcal{L}\|^2 C_{\bar{x}} + N\theta_{\max} C_\eta\right]} \\ &= C_e. \end{aligned} \tag{37}$$

Since the estimation error $\|e_i(t)\|$ is piecewise continuous differentiable for $t \in [t_k^i, t_{k+1}^i)$, the Dini derivative of estimation error $\|e_i(t)\|$ is as follows:

$$\begin{aligned} D^+\|\dot{e}_i(t)\| = \frac{e_i(t)\dot{e}_i(t)}{\|e_i(t)\|} &\le \|\dot{e}_i(t)\| \\ &\le \|A\|\|e_i(t)\| + \|B\|\|\mathrm{sat}(u_i(t))\| \\ &\le \|A\|C_e| + \|BK\||\mathcal{N}_i|u_{\max} \\ &= \hat{C}_e. \end{aligned} \tag{38}$$

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

The event is only triggered when the condition in (6) is satisfied. Thus, just before the next triggering instant, *i.e.*, $t = t_k^{i-}$, we have

$$\left\|e_i\left(t_k^{i-}\right)\right\| \geq \sqrt{\frac{\left\|\alpha\sum_{j\in\mathcal{N}_i}\left(\hat{x}_i\left(t_k^{i-}\right)-\hat{x}_j\left(t_k^{i-}\right)\right)\right\|^2+\theta_i\eta_i\left(t_k^{i-}\right)}{\lambda_{\max}(\Omega)}} \geq \sqrt{\frac{\theta_i\eta_i\left(t_k^{i-}\right)}{\lambda_{\max}(\Omega)}} \geq \sqrt{\frac{\theta_i\eta_i(0)}{\lambda_{\max}(\Omega)}}e^{-\frac{1}{2}(\theta_i+\beta_i)t_k^{i-}}. \tag{39}$$

Suppose that $t^i_{N(\epsilon)}$ is the $N(\epsilon)$-th triggering instant, and $t^i_{N(\epsilon)+1}$ is the $(N(\epsilon)+1)$-th triggering instant. Then it follows from (38) and (39) that:

$$\begin{aligned} t^i_{N(\epsilon)+1}-t^i_{N(\epsilon)} &> t^{i-}_{N(\epsilon)+1}-t^i_{N(\epsilon)} \\ &\geq \frac{1}{\hat{C}_e}\sqrt{\frac{\theta_i\eta_i(0)}{\lambda_{\max}(\Omega)}}e^{-\frac{1}{2}(\theta_i+\beta_i)t^{i-}_{N(\epsilon)+1}} \\ &\geq \frac{1}{\hat{C}_e}\sqrt{\frac{\theta_i\eta_i(0)}{\lambda_{\max}(\Omega)}}e^{-\frac{1}{2}(\theta_i+\beta_i)(T_0+\epsilon)}. \end{aligned} \tag{40}$$

Let ϵ is the solution of the following:

$$\frac{1}{\hat{C}_e}\sqrt{\frac{\theta_i\eta_i(0)}{\lambda_{\max}(\Omega)}}e^{-\frac{1}{2}(\theta_i+\beta_i)T_0} = 2\epsilon e^{\frac{1}{2}(\theta_i+\beta_i)\epsilon}. \tag{41}$$

Thus it follows directly from (40) and (41) that

$$t^i_{N(\epsilon)+1}-t^i_{N(\epsilon)} \geq 2\epsilon \tag{42}$$

which violates the supposition that $t^i_{N(\epsilon)+1}-t^i_{N(\epsilon)} < 2\epsilon$. Thus, Zeno behavior doesn't exist for agent i. The proof is completed. ■

Remark 5: For the DETM system, it is crucial to design a suitable threshold condition that reduces data transmission while simultaneously preventing the occurrence of Zeno behavior. Zeno behavior means that the events can occur indefinitely in finite amount of time, leading to continuous transmission between agents and consuming too much communication resource. Thus, the elimination of Zeno behavior is a fundamental reqiurement for implementing DETM. In practical applications, the DETM and the designed control strategy should be operated in discrete-time manner due to the implementation of digital platforms. To achieve this, Euler method can be adopted to discretize DETM and control strategy. The discretization form of DETC can be obtained as follows: $t^i_{k+1}h = t_kh + \min_{l\geq 1}\{lh| - \alpha\|\sum_{j\in\mathcal{M}_i}\pi_{ij}(\hat{x}_i(t^i_k+lh)-\hat{x}_j(t^i_k+lh))\|^2 + e_i^{\mathrm{T}}(t^i_k+lh)\Omega e_i(t^i_k+lh) \geq \eta_i(t^i_k+lh)\}$, the evolution of the internal variable η_i is given as $\eta_i((k+1)h) = (1-\beta_ih)\eta_i(kh) + \alpha h\|\sum_{j\in\mathcal{M}_i}\pi_{ij}(\hat{x}_i(kh)-\hat{x}_j(kh))\|^2 + e_i^{\mathrm{T}}(kh)\Omega e_i(kh)$, where h is sampling period.

Remark 6: The correlation of design parameters, DoS attacks and system performance is revealed according to the derived conditions in Theorem 1. The inequality (18) denotes the resilient level of the system regarding DoS attacks. c_1 and c_2 represent the convergence rate when the system is under control and divergence rate when the system is subject to attacks, respectively. It can be seen that if there are larger value of c_1 and smaller value of c_2, then the tolerance of the system regarding DoS attacks can be improved. Under the condition that the LMIs are solvable, maximize the convergence rate and minimize the divergence rate as much as possible, as this will help enhance the supported frequency and duration. If an attacker uses values greater than what the system can tolerate, it may lead to destabilization or a loss of consensus within the system. Additionally, the value of c_1 depends on the minimum between γ_1 and β_i. The choice of γ_1 directly affects the feasibility of (19) which affects the design of controller. β_i makes a difference to DETM, the results of which can be reflected in the number of triggers and system performance. Based on the above analysis, there is a tradeoff between convergence rate and triggering times.

Note that in Theorem 1, the feedback gain K and the positive matrix Ω cannot be directly solved by (19) due to the coupling terms and complicated computations. Then an effective method approach is proposed, we can obtain the following design criterion by utilizing congruent transformation.

Theorem 2: Consider a MAS (1) subject to DoS attacks and input saturation, and it satisfies Assumptions 1 and 2. For given $\theta_i > 0$, $\beta_i > 0$, $\gamma_1 > 0$, $c_2 > 0$, $\alpha > 0$, the consensus problem can be realized under the DETM (6) and state-prediction-based controller (9) via unreliable and limited communication networks, if DoS attacks satisfy (18), and if there exist a scalar $a > 0$, positive-definite matrices X, $\bar{\Omega}$ and $\tilde{\Upsilon}$, and $\tilde{\mathscr{W}}$ and Y such that the following inequalities hold:

$$\begin{bmatrix} \hat{\Sigma}_{11} & 0 & \hat{\Sigma}_{13} & 0 & \hat{\Sigma}_{15} & \hat{\Sigma}_{16} \\ * & -2\tilde{\Upsilon} & 0 & 0 & 0 & 0 \\ * & * & \hat{\Sigma}_{33} & 0 & \hat{\Sigma}_{35} & 0 \\ * & * & * & -\bar{\Omega} & 0 & 0 \\ * & * & * & * & \hat{\Sigma}_{55} & \hat{\Sigma}_{56} \\ * & * & * & * & * & \hat{\Sigma}_{66} \end{bmatrix} < 0 \tag{43}$$

$$\begin{bmatrix} I_N\otimes X & I_N\otimes\tilde{\mathscr{W}}^{\mathrm{T}} \\ * & \varepsilon^{-1}u_{\max}^2 \end{bmatrix} \geq 0 \tag{44}$$

$$\begin{bmatrix} I_N\otimes\bar{\Omega} & \tilde{\mathcal{J}}\otimes X \\ * & \frac{1}{\alpha(1+a)}I_{Nn_x} \end{bmatrix} > 0 \tag{45}$$

where $\hat{\Sigma}_{11} = I_{N-1}\otimes(AX+XA^{\mathrm{T}}) - 2\mathcal{J}\otimes BY + \gamma_1(I_{N-1}\otimes X)$, $\hat{\Sigma}_{13} = -\mathcal{J}_N\otimes B\tilde{\Upsilon} + \mathcal{J}\otimes Y^{\mathrm{T}} - I_{N-1}\otimes\tilde{\mathscr{W}}^{\mathrm{T}}$, $\hat{\Sigma}_{15} = -\mathcal{J}\otimes BY$, $\hat{\Sigma}_{16} = \mathcal{J}\otimes X$, $\hat{\Sigma}_{33} = -2I_{N-1}\otimes\tilde{\Upsilon}$, $\hat{\Sigma}_{35} = \mathcal{J}\otimes Y$, $\hat{\Sigma}_{55} = -I_{N-1}\otimes\bar{\Omega}$, $\hat{\Sigma}_{56} = \mathcal{J}\otimes X$. And the feedback gain can be computed as $K = YX^{-1}$.

Proof: Pre- and post-multiplying inequalities (19), (20), and (21) with $\mathrm{diag}\{I_{N-1}\otimes P^{-1}, \Upsilon^{-1}, I_{N-1}\otimes\Upsilon^{-1}, P^{-1}, I_{N-1}\otimes P^{-1}, I_{(N-1)n_x}\}$, $\mathrm{diag}\{I_N\otimes P^{-1}, I_{Nn_x}\}$, $\mathrm{diag}\{I_N\otimes P^{-1}, I_{Nn_x}\}$, respectively. Define $X = P^{-1}$, $Y = KX$, $\tilde{\Upsilon} = \Upsilon^{-1}$, $X\Omega X = \bar{\Omega}$, $\tilde{\mathscr{W}} = \mathscr{W}X$. Then, we can get the inequalities (43)–(45). ■

Remark 7: The eigenvalues of the Laplacian matrix is required in Theorems 1 and 2. This implies the necessity of the global information of graph. In practice, this information can be obtained in certain situations. Therefore, this requirement is practically acceptable.

In Theorem 2, the design criteria terms of LMIs are provided to achieve consensus for MAS with state-prediction-based DETM control protocol under DoS attacks and input

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

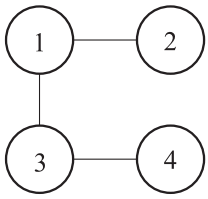


Fig. 2. Communication topology of four agents.

saturation. In the next section, a simulation is given to verify the effectiveness the proposed controller.

IV. Simulation Example

A simulation is given to illustrate the theoretical results. Consider a 4-agent MAS with the communication network $\mathcal{G}$ as plotted in Fig. 2, the dynamics of each agent is given by

$$\dot{x}_i(t) = \begin{bmatrix} -2 & 1 \\ 0.1 & 0.2 \end{bmatrix} x_i(t) + \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix} \text{sat}(u_i(t)).$$

According to the communication graph of 4 agents in Fig. 2, the Laplacian matrix $\mathcal{L}$ is

$$\mathcal{L} = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

The eigenvalues of $\mathcal{L}$ are $[0, 0.5858, 2.000, 3.4142]^{\mathrm{T}}$. The initial internal states $\eta_i(0)$ of the DETM are set as $(10, 20, 30, 40)$. The parameters of DETM are set as $\theta_i = 30$, $\beta_i = 0.5$, $\alpha = 0.07$, respectively. The saturation bound $u_{\max} = 2$, $\varepsilon = 0.5$,$\gamma_1 = 5$, $c_2 = 0.5$. By working out the LMIs in Theorem 2, we get the feedback gain is computed as $K = \begin{bmatrix} -22.5303 & 69.0438 \end{bmatrix}$,

$$P = \begin{bmatrix} 1.9652 & -3.8807 \\ -3.8807 & 8.5740 \end{bmatrix}, Q = \begin{bmatrix} 0.7064 & -0.2577 \\ -0.2577 & 1.1827 \end{bmatrix}.$$

Then the parameters are calculated as $\lambda_1 = 7.5039$, $\lambda_2 = 17.4620$. Note that the limitations on frequency and duration of DoS attacks in Assumptions 1 and 2 should satisfy the inequalities (18). Accordingly, we have $T_2 > 11$, $F_2 > 12$. The regularization terms of DoS attacks are given as $T_1 = 4$, $F_1 = 8$.

To demonstrate the virtue of the designed proposed control protocol for MASs subject to DoS attacks and input saturation, the simulation results are depicted in Figs. 3–6. The initial states of each agent are chosen within the set $\psi(P, \varepsilon)$ as $x(0) = [1.45, 1.45, 1.67, 1.67, 1.28, 1.28, 1.58, 1.58]^{\mathrm{T}}$.

Fig. 3 plots the state evolution of each agent under aperiodic DoS attacks and input saturation. It can be observed that during the DoS attack period, the states of agents appear to diverge due to the absence of neighbors' information and control input. Fig. 4 plots the evolution of saturated control input. With the control effect, the agents gradually converge to the same state, and the required value of control input becomes smaller and could not be saturated. Combined with Figs. 3 and 4, it can be found that although the consensus cannot be achieved at first because of the saturated control input $\text{sat}(u_i(t))$ and DoS

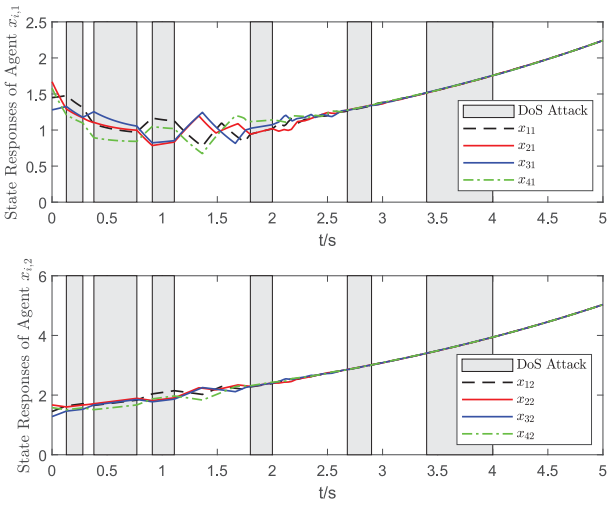


Fig. 3. State trajectories of $x_{i,1}$, $x_{i,2}$ subject to input saturation and DoS attacks.

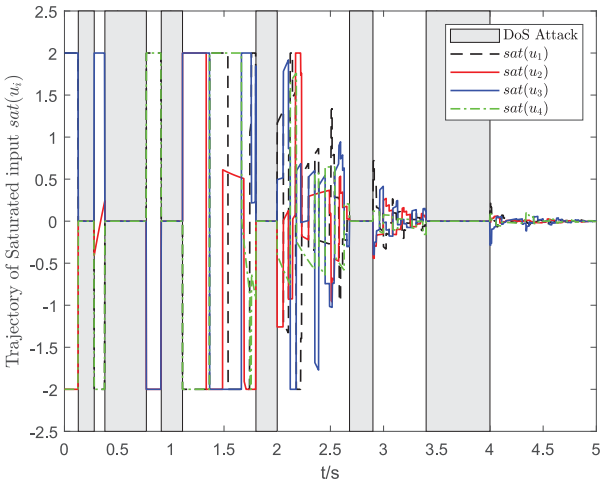


Fig. 4. Saturated control input $\text{sat}(u_i)$.

attacks. The agents tend to reach consensus with the effective control strategy, which verifies the validity of our proposed controller.

Note that the most important problem to implement DETM is to exclude Zeno behavior, the continuous communication between agents can be avoided. From Fig. 5, the interevent release intervals are shown and there is no Zeno behavior. The length of blue vertical lines in Fig. 5 denotes the elapse time between adjacent triggering instants. Moreover, the trajectories of different internal dynamic states are described in Fig. 6. It's observed that internal variable η_i is non-monotonicity. Once the DoS attacks occur, the value of $\dot{\eta}_i$ is held to zero and η_i is greater than zero during the communication intervals. Thus, $\eta_i(t) \geq 0$ can be ensured.

V. Conclusion

The secure consensus problem have been investigated for MASs with limited-capacity actuators, operating through unreliable communication networks. These systems experience

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

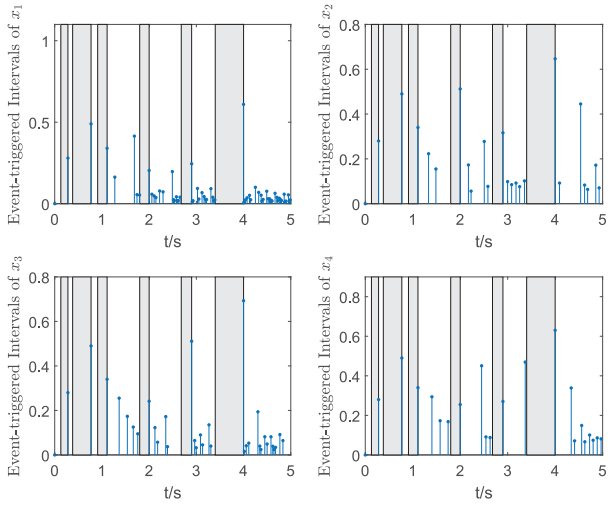


Fig. 5. Event release intervals of each agent with DETM.

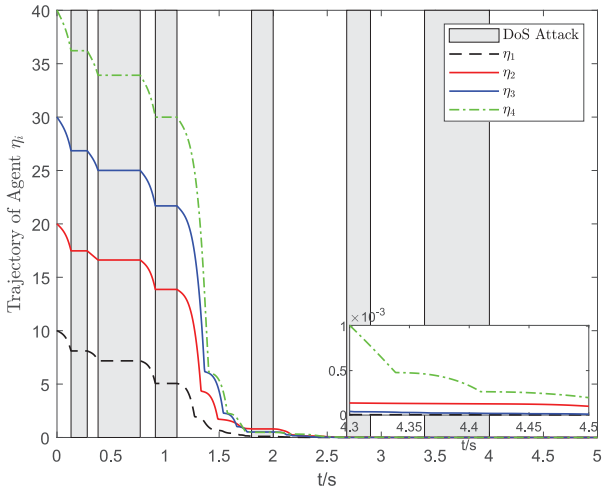


Fig. 6. Evolution of internal state η_i.

sector-bounded input saturation in their control inputs and are subject to aperiodic DoS attacks. To address this issues, a state-prediction-based DETM has been proposed for determining the communication instants between adjacent agents. Thus, the requirements of continuous monitoring of agents' states and continuous communication between agents are no longer needed, resulting in significant energy saving, which is crucial in some battery-powered scenarios. Sufficient conditions have been given to assure the consensus of MASs. The correlation between consensus, the DoS attacks and input saturation bound has been revealed. Furthermore, the Zeno behavior is proved to be nonexistent. Finally, a simulation has been conducted to illustrate the effectiveness of the proposed method. In the future, we will further investigate the secure control problems of MASs suffering from complex attacks, and we will consider a more general setting, in which MASs are heterogeneous and the communication topology is time-varying. These factors will significantly increase the difficulty of the research. However, these issues are well worth considering. Besides, future research will also explore event-triggered encoding and decoding strategies to ensure data integrity and privacy of communication between agents under complex attacks.

REFERENCES

[1] C. Liu, B. Jiang, X. Wang, Y. Zhang, and S. Xie, "Event-based distributed secure control of unmanned surface vehicles with DoS attacks," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 54, no. 4, pp. 2159–2170, Apr. 2024.

[2] X. Ge, Q.-L. Han, and Z. Wang, "A dynamic event-triggered transmission scheme for distributed set-membership estimation over wireless sensor networks," *IEEE Trans. Cybern.*, vol. 49, no. 1, pp. 171–183, Jan. 2019.

[3] Y. Izmirlioglu, L. Pham, T. C. Son, and E. Pontelli, "A survey of multi-agent systems for smartgrids," *Energies*, vol. 17, no. 15, p. 3620, 2024.

[4] X. He, Z. Wang, C. Gao, and D. Zhou, "Consensus control for multiagent systems under asymmetric actuator saturations with applications to mobile train lifting Jack systems," *IEEE Trans. Ind. Informat.*, vol. 19, no. 10, pp. 10224–10232, Oct. 2023.

[5] R. Olfati-Saber and R. M. Murray, "Consensus problems in networks of agents with switching topology and time-delays," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1520–1533, Sep. 2004.

[6] W. Chen, D. Ding, H. Dong, and G. Wei, "Distributed resilient filtering for power systems subject to denial-of-service attacks," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 49, no. 8, pp. 1688–1697, Aug. 2019.

[7] Y. Yuan, H. Yuan, L. Guo, H. Yang, and S. Sun, "Resilient control of networked control system under DoS attacks: A unified game approach," *IEEE Trans. Ind. Informat.*, vol. 12, no. 5, pp. 1786–1794, Oct. 2016.

[8] C. D. Persis and P. Tesi, "Input-to-state stabilizing control under denial-of-service," *IEEE Trans. Autom. Control*, vol. 60, no. 11, pp. 2930–2944, Nov. 2015.

[9] W. Xu, G. Hu, D. W. C. Ho, and Z. Feng, "Distributed secure cooperative control under denial-of-service attacks from multiple adversaries," *IEEE Trans. Cybern.*, vol. 50, no. 8, pp. 3458–3467, Aug. 2020.

[10] Y.-C. Sun and G.-H. Yang, "Event-triggered distributed state estimation for multiagent systems under DoS attacks," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 6901–6910, Jul. 2022.

[11] D. Zhang and G. Feng, "A new switched system approach to leader–follower consensus of heterogeneous linear multiagent systems with DoS attack," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 51, no. 2, pp. 1258–1266, Feb. 2021.

[12] Y. Yang, Y. Li, D. Yue, Y.-C. Tian, and X. Ding, "Distributed secure consensus control with event-triggering for multiagent systems under DoS attacks," *IEEE Trans. Cybern.*, vol. 51, no. 6, pp. 2916–2928, Jun. 2021.

[13] M. Xing, J. Lu, Y. Liu, and X. Chen, "Event-based bipartite consensus of multi-agent systems subject to DoS attacks," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 1, pp. 68–80, Jan./Feb. 2023.

[14] C. Deng and G.-H. Yang, "Consensus of linear multiagent systems with actuator saturation and external disturbances," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 64, no. 3, pp. 284–288, Mar. 2017.

[15] C. Gao, Z. Wang, X. He, and Q.-L. Han, "Consensus control of linear multiagent systems under actuator imperfection: When saturation meets fault," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 4, pp. 2651–2663, Apr. 2022.

[16] Z. Li and J. Zhao, "Adaptive consensus of non-strict feedback switched multi-agent systems with input saturations," *IEEE/CAA J. Automatica Sinica*, vol. 8, no. 11, pp. 1752–1761, Nov. 2021.

[17] Y. Zou, Z. Zuo, K. Xia, and Z. Ding, "Consensus of saturated multiagent systems: Tolerance to nonidentical asymmetric saturation levels," *IEEE Trans. Autom. Control*, vol. 69, no. 3, pp. 1804–1811, Mar. 2024.

[18] S. Du, Q. Yan, L. Dong, and J. Qiao, "Secure consensus of multiagent systems with input saturation and distributed multiple DoS attacks," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 4, pp. 2246–2250, Apr. 2022.

[19] A. Hu, J. Cao, M. Hu, and L. Guo, "Event-triggered group consensus for multi-agent systems subject to input saturation," *J. Frankl. Inst.*, vol. 355, no. 15, pp. 7384–7400, 2018.

[20] H. Wu and H. Su, "Observer-based consensus for positive multiagent systems with directed topology and nonlinear control input," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 49, no. 7, pp. 1459–1469, Jul. 2019.

[21] Z. Zuo, J. Ji, Z. Zhang, Y. Wang, and W. Zhang, "Consensus of multi-agent systems with asymmetric input saturation over directed graph," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 4, pp. 1515–1519, Apr. 2023.

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

[22] D. V. Dimarogonas, E. Frazzoli, and K. H. Johansson, "Distributed event-triggered control for multi-agent systems," *IEEE Trans. Autom. Control*, vol. 57, no. 5, pp. 1291–1297, May 2012.
[23] P. Tabuada, "Event-triggered real-time scheduling of stabilizing control tasks," *IEEE Trans. Autom. Control*, vol. 52, no. 9, pp. 1680–1685, Sep. 2007.
[24] A. Girard, "Dynamic triggering mechanisms for event-triggered control," *IEEE Trans. Autom. Control*, vol. 60, no. 7, pp. 1992–1997, Jul. 2015.
[25] Z. Cheng, D. Yue, S. Hu, C. Huang, C. Dou, and X. Ding, "Resilient dynamic event-triggered control for multi-area power systems with renewable energy penetration under DoS attacks," *IET Control Theory Appl.*, vol. 14, no. 16, pp. 2267–2279, 2020.
[26] Y. Cui, H. Sun, L. Hou, and K. Shi, "Dynamic event-triggered adaptive neural network Decentralized output-feedback control for nonlinear interconnected systems with hybrid cyber attacks and its application," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 54, no. 4, pp. 2149–2158, Apr. 2024.
[27] A. Amini, A. Asif, and A. Mohammadi, "A unified optimization for resilient dynamic event-triggering consensus under denial of service," *IEEE Trans. Cybern.*, vol. 52, no. 5, pp. 2872–2884, May 2022.
[28] X. Yi, K. Liu, D. V. Dimarogonas, and K. H. Johansson, "Distributed dynamic event-triggered control for multi-agent systems," in *Proc. IEEE 56th Annu. Conf. Decis. Control (CDC)*, 2017, pp. 6683–6698.
[29] W. Hu, C. Yang, T. Huang, and W. Gui, "A distributed dynamic event-triggered control approach to consensus of linear multiagent systems with directed networks," *IEEE Trans. Cybern.*, vol. 50, no. 2, pp. 869–874, Feb. 2020.
[30] W. Song, J. Feng, H. Zhang, and W. Wang, "Dynamic event-triggered formation control for heterogeneous multiagent systems with nonautonomous leader agent," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 9685–9699, Dec. 2023.
[31] D. Yang, W. Ren, X. Liu, and W. Chen, "Decentralized event-triggered consensus for linear multi-agent systems under general directed graphs," *Automatica*, vol. 69, pp. 242–249, Jul. 2016.
[32] C.-Q. Ma and J.-F. Zhang, "Necessary and sufficient conditions for consensusability of linear multi-agent systems," *IEEE Trans. Autom. Control*, vol. 55, no. 5, pp. 1263–1268, May 2010.
[33] S. Tarbouriech, G. Garcia , J. M. Gomes da Silva Jr., and I. Queinnec, *Stability and Stabilization of Linear Systems With Saturating Actuators*. London, U.K.: Springer, 2011.
[34] Y. Xu, M. Fang, P. Shi, and Z.-G. Wu, "Event-based secure consensus of mutiagent systems against DoS attacks," *IEEE Trans. Cybern.*, vol. 50, no. 8, pp. 3468–3476, Aug. 2020.
[35] N. Zhao, P. Shi, and X. Wen, "Dynamic event-triggered approach for networked control systems under denial of service attacks," *Int. J. Robust Nonlin. Control*, vol. 31, no. 5, pp. 1774–1795, 2021.
[36] S. Hu, D. Yue, X. Xie, X. Chen, and X. Yin, "Resilient event-triggered controller synthesis of networked control systems under periodic DoS jamming attacks," *IEEE Trans. Cybern.*, vol. 49, no. 12, pp. 4271–4281, Dec. 2019.
[37] Y. Xu, M. Fang, Z.-G. Wu, Y.-J. Pan, M. Chadli, and T. Huang, "Input-based event-triggering consensus of multiagent systems under denial-of-service attacks," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 4, pp. 1455–1464, Apr. 2020.



Yifang Zhang was born in 1996. She received the B.S. degree in automation from Chongqing University, Chongqing, China, in 2018, and the Ph.D. degree in control science and engineering from Zhejiang University, Hangzhou, China, in 2023.

She is currently a Postdoctoral Fellow with the Department of Mechanical Engineering, The University of Hong Kong, Hong Kong. Her current research interests include security of cyber–physical system, multiagent system, and Markov jump systems.



Zheng-Guang Wu was born in 1982. He received the B.S. and M.S. degrees in mathematics from Zhejiang Normal University, Jinhua, China, in 2004 and 2007, respectively, and the Ph.D. degree in control science and engineering from Zhejiang University, Hangzhou, China, in 2011.

He is currently with the Institute of Cyber-Systems and Control, Zhejiang University. His current research interests include hybrid systems, networked systems. and computational intelligence.



Ka-Wai Kwok (Senior Member, IEEE) received the B.Eng. and M.Phil. degrees from the Department of Automation and Computer-Aided Engineering [currently Mechanical and Automation Engineering (MAE)], The Chinese University of Hong Kong (CUHK), Hong Kong, in 2003 and 2005, respectively, and the Ph.D. degree from the Hamlyn Centre for Robotic Surgery, Department of Computing, Imperial College London, London, U.K., in 2012.

After then, he was awarded the Croucher Foundation Fellowship 2013, which supported his research jointly supervised by Advisors with The University of Georgia, Athens, GA, USA; and Brigham and Women's Hospital, Boston, MA, USA; and Harvard Medical School, Boston, MA, USA. He is currently a Professor with MAE, CUHK. Before then, he served as a Faculty with the Department of Mechanical Engineering, The University of Hong Kong, Hong Kong, from 2014 to 2024. He is also a Co-Founder and the Director of Agilis Robotics Ltd., Hong Kong, aiming at advancing the interventional endoscopy with small, flexible robotic instruments, and their intelligent control systems. He has participated in various designs of robotic devices/interfaces for endoscopy and MRI-guided interventions. To date, he has co-authored >165 peer-reviewed articles with >80 Clinical Fellows and >160 Scientists/Engineers. His research interests focus on surgical robotics, intraoperative image processing, and their uses of control and intelligent systems.

Dr. Kwok's multidisciplinary work has been recognized with various (>10) awards in international conferences/journals, such as the largest flagship conferences of robotics, ICRA, and IROS. He was the recipient of the ICRA Best Conference Paper Award in 2018, and the IROS Toshio Fukuda Young Professional Award in 2020. He also obtained awards in his early career for robotics, such as the Early Career Awards 2015/2016 offered by Research Grants Council of Hong Kong, the Actuators 2020 Young Investigator Award, the HKU 2019 and 2020 Outstanding Young Researcher Award, the HKU Young Innovator Award 2020, and the HKU Research Output Prize 2021 and 2022. He serves with editor boards for IROS 2017–22, ICRA 2019–23, *IEEE Robotics and Automation Magazine*, as well as *Annals of Biomedical Engineering* and *Proceedings of the Institution of Mechanical Engineers—Part I: Journal of Systems and Control Engineering*. He is the Principal Investigator of Group for Interventional Robotic and Imaging Systems, which has (>5) inventions licensed/transferred from University to industry in support for their commercialization.



Tingwen Huang (Fellow, IEEE) received the B.S. degree in mathematics from Southwest Normal University (now Southwest University), Chongqing, China, in 1990, the M.S. degree in applied mathematics from Sichuan University, Chengdu, China, in 1993, and the Ph.D. degree in applied mathematics from Texas A&M University, College Station, TX, USA, in 2002.

He was a Visiting Assistant Professor with Texas A&M University. He joined the Texas A&M University at Qatar, Doha, Qatar, as an Assistant Professor, in August 2003, where he was promoted to a Professor in 2013. His research interests include neural networks, computational intelligence, smart grid, chaotic dynamical systems, complex networks, and optimization and control.

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.



Prasun Chakrabarti received the D.Sc h.c., D.Eng. h.c., and Ph.D. h.c. degrees, the Ph.D. (Engg.) degree in computer science and engineering from Jadavpur University, Kolkata, India, in 2009, 2021, and 2021, respectively, and the D.Litt. degree in management from Sambalpur University, Sambalpur, India, in 2022.

He is serving as the Deputy Provost (Deputy Vice Chancellor), ITM SLS Baroda University, Vadodara, India. He has supervised 11 PhDs successfully and also has 100+ Scopus/SCIE publications, 55 granted international patents, 13 granted Indian copyrights and 11 books in his credit. He has been Visiting Professor with the Graduate School of IPS, Waseda University, Shinjuku, Japan in 2012; University of Mauritius, Moka, Mauritius, in 2015; Lincoln University College Malaysia, Kota Bharu, Malaysia, in 2018; Nanyang Technological University, Singapore in 2015, 2016, and 2019; and a Academic Visitor, Asian Institute of Technology, Khlong Nueng, Thailand, in 2019; and TLIAP-National University of Singapore, Singapore in 2019. Currently he is an Adjunct Professor and a Scientist of Wales Institute of Digital Information, Cardiff, U.K.; Don State Technical University, Rostov-on-Don, Russia; Shiraz University of Medical Sciences, Shiraz, Iran; and Scientific Innovation Research Group, Egypt.

Dr. Chakrabarti is Fellow of IET (UK), IEI, IETE, INSS and former Fellow of Royal Society of Arts London.



Le Zhou (Senior Member, IEEE) received the B.Eng. and Ph.D. degrees from the Department of Control Science and Engineering, Zhejiang University, Hangzhou, China, in 2010 and 2015, respectively.

He was a Visiting Scholar with the Viterbi School of Engineering, University of Southern California, Los Angeles, CA, USA, from December 2013 to June 2014, and a Visiting Scholar with the Department of Chemical Engineering, National Tsinghua University, Hsinchu, Taiwan, from January 2019 to March 2019. He is currently a Professor with the School of Automation and Electrical Engineering, Zhejiang University of Science and Technology, Hangzhou. His research interests include industrial process modeling, monitoring and fault diagnosis, soft sensor modeling, and deep learning.

Authorized licensed use limited to: Chinese University of Hong Kong. Downloaded on June 06,2025 at 03:02:59 UTC from IEEE Xplore. Restrictions apply.